(iv) Truth table for $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$

| $p$ | $q$ | $r$ | $\neg p$ | $(\neg p \rightarrow r)$ | $p \leftrightarrow q$ | $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$ |
|---|---|---|---|---|---|---|
| T | T | T | F | T | T | T |
| T | T | F | F | T | T | T |
| T | F | T | F | T | F | F |
| T | F | F | F | T | F | F |
| F | T | T | T | T | F | F |
| F | T | F | T | F | F | F |
| F | F | T | T | T | T | T |
| F | F | F | T | F | T | F |

7. (i) Contradiction (ii) Tautology (iii) Contingency (iv) Tautology

12. $p \rightarrow (q \rightarrow p)$ is a Tautology.

13. Yes. The statements are logically equivalent.

## Exercise 12.3

**Choose the appropriate answer from the given distractors.**

| Q | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| A | (2) | (3) | (2) | (4) | (2) | (2) | (3) | (4) | (3) | (2) |
| Q | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | (4) | (1) | (3) | (3) | (3) | (2) | (4) | (3) | (1) | (4) |



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